A FINITE TYPE FLAT MODULE WITH FULL SUPPORT OVER A LOCALLY FINITELY PRESENTED MORPHISM MAKES IT UNIVERSALLY OPEN

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is a presentation reduction with the full-support hypothesis carried along; the hypothesis X = Supp(F) is part of the counted statement span.

Editorial note. The single reference of the counted spans is recorded as a non-load-bearing navigation pointer; the argument that consumes it stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- Reduction of universal openness to a presentation. used at proof lines 6549-6553. Replaces the algebraic space statement by a scheme statement.
- The scheme case: a finite type flat module with full support gives universal openness. Morphisms, Lemma morphisms-lemma-pf-flat-module-open, used at proof lines 6554-6555. Is the input that closes the argument.
- Surjective etale presentations fitting into a square. used at proof lines 6550-6552. Produces V and U.

Source reference. Original lemma 0CVT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 16148–16148. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module. Assume f locally finite presentation, $\mathcal{F}$ of finite type, $X = Supp(\mathcal{F})$, and $\mathcal{F}$ flat over Y. Then f is universally open.*

Proof. Choose a surjective étale morphism $\varphi : V \to Y$ where V is a scheme. Choose a surjective étale morphism $U \to V \times_Y X$ where U is a scheme. Then it suffices to prove the lemma for $U \to V$ and the quasi-coherent $\mathcal{O}_V$-module $\varphi^*\mathcal{F}$. Hence this lemma follows from the case of schemes, see Morphisms, Lemma 0CVT. □

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.